

Soumen Kumar

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Education

Indian Institute of Technology, Jodhpur

October 2022 - May 2026

BTech in Engineering Science(Major-Computer Science and Engineering) CGPA - 7.60

Jodhpur, Rajasthan

Relevant Coursework

- | | | | |
|-----------------------|--------------------------|---------------------|---------------------------|
| • Data Structures | • Database Management | • Operating Systems | • Computer Networks |
| • Machine Learning | • Cybersecurity | • Cloud Computing | • Intro to CS and OOPS |
| • Algorithms Analysis | • Probability Statistics | • Cryptography | • Engineering Maths(I/II) |

Experience

CybraneX

November 2024 - April 2026

SDE Intern

Remote

- Developed **CUDA-Q-based Quantum ML** pipelines with hybrid classical-quantum optimization for scheduling and logistics, achieving up to **30% faster convergence**.
- Built **full-stack platforms (FastAPI + Next.js)** enabling remote access to **physics instrumentation** (e.g., NMR) with scalable APIs and real-time data pipelines.

InfoEdge (naukri.com)

May 2025 - July 2025

Data Science Intern

Noida, India

- Revamped the **Minis recommendation system** using **Bayesian statistics**, **LambdaRank**, and **XGBoost** with **PySpark**, **SQL**, and **Datalake**, boosting personalization accuracy by **3%**.
- Built a scalable **analytics dashboard** using **Pandas** and **Streamlit** for real-time insights on Minis, and developed an **LLM-powered article generator** with **FastAPI** integrated with **LLaMA**, **Qwen**, and **Flux Schnell**, producing **PIL outputs within 5s**.

Projects

Intelligent Document Analyst🤖 | Hackathon

July 2025 – August 2025

- Built a **persona-aware document analysis system** using **MiniLM**, extracting key sections from 3–10 PDFs fully **offline on CPU**.
- Developed a **Dockerized pipeline** for parsing, semantic ranking, and JSON output in under **60 seconds**.
- Enabled adaptive section ranking for diverse domains including research, business, and education.

Cancer Diagnostic App🤖 | Project under (IIT'J-AIIMS'J)

October 2024 – January 2025

- Created a mobile diagnostic tool for NSCLC patients using **Kotlin** and **XML**, integrating **NCCN guidelines** for real-time, personalized cancer diagnosis and treatment planning.
- Integrated **Firebase Authentication**, **Firestore DB**, and **Firebase Storage** to handle sensitive medical data securely, with a scalable architecture supporting future expansion to 5+ cancer types.

MedSync - Pharmacy Management Platform🤖 | Course Project

September 2024 – November 2024

- Built a full-stack system linking 500+ users to IITJ PHC Pharmacy for live medicine browsing, order tracking, and real-time notifications using **React**, **Firebase**, and **Google Cloud**.
- Launched a feature-rich admin panel with stock alerts, UPI payment support, and user analytics, improving operational turnaround time by 30% and enabling seamless communication with customers.

Technical Skills

Languages:Python,C/C++,Shell Scripting, HTML/CSS, SQL, Kotlin, XML

Developer Tools:Git,Firebase, Android Studio,WinAPI,Azure,Mongo DB,ElasticSearch

Technologies/Frameworks: Linux, React.js, Scikit-Learn, OpenCV, Pandas, Scipy,PyTorch

Achievements

- Expert@CodeForces(Rating-1864)
- Department Rank 5 in B.Tech CS+EE
- Got 3rd Prize(Project Demonstration) in Industry Day-2024@IITJ.
- Part of IITJ Team in ISRO-URC 24

Positions of Responsibility

- Development Head @ Akshar IITJ
- Secretary @ ES Board of Departmental Societies
- Team Captain@ Quiz Society IITJ